Lesson 5.2 - Probability Formulas

> **Video:** Probability Formulas, Symbols & Notations - Marginal, Joint, & Conditional Probabilities by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

This lesson collects the main probability formulas and symbols in one place. All examples use this setup:

- Sample space $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $A = \{1, 2, 3, 4\}, \quad B = \{6, 7, 8\}, \quad C = \{1, 7, 8, 9\}$

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1. Marginal probability - a single event

The probability of **one** event, regardless of any others:

$$P(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

- $P(A) = \frac{4}{9} \approx 0.444$ (44.4%)
- $P(B) = \frac{3}{9} = \frac{1}{3} \approx 0.333$ (33.3%)
- $P(C) = \frac{4}{9} \approx 0.444$ (44.4%)

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2. Union and intersection symbols

Symbol	Name	Meaning	Example
$A \cup B$	Union	Everything in A **or** B (or both), listed once	$B \cup C = \{1, 6, 7, 8, 9\}$
$A \cap B$	Intersection	Only what is in **both** A **and** B	$B \cap C = \{7, 8\}$

$A \cap B = \varnothing$ (no shared elements), so A and B are **mutually exclusive**.

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3. Union probability - $P(A \cup B)$ (addition rule)

$P(A \cup B)$ is read "the probability of A **or** B" (at least one of them happens).

$$\begin{aligned} \text{Words:} \quad & P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) \\ \text{Symbols:} \quad & P(A \cup B) = P(A) + P(B) - P(A \cap B) \end{aligned}$$

The subtraction stops the overlap $A \cap B$ from being counted **twice**.

Mutually exclusive events can't happen together, so $A \cap B = \varnothing$ and $P(A \cap B) = 0$:

$$\begin{aligned} \text{Words:} \quad & P(A \text{ or } B) = P(A) + P(B) \\ \text{Symbols:} \quad & P(A \cup B) = P(A) + P(B) \end{aligned}$$

Example - $A \cup B$ (mutually exclusive): $A \cup B = \{1, 2, 3, 4, 6, 7, 8\}$

$$P(A \cup B) = \frac{7}{9} \approx 0.778 \quad \left(= \tfrac{4}{9} + \tfrac{3}{9}\right)$$

Example - $B \cup C$ (they overlap at 7 and 8): $B \cap C = \{7, 8\}$, so $P(B \cap C) = \frac{2}{9}$

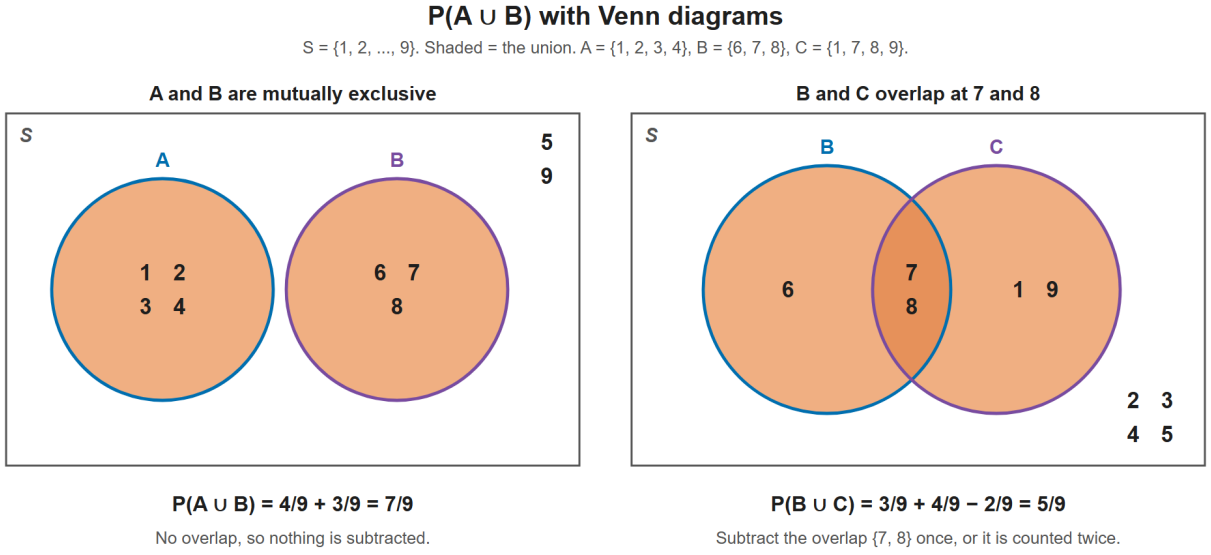
$$P(B \cup C) = P(B) + P(C) - P(B \cap C) = \frac{3}{9} + \frac{4}{9} - \frac{2}{9} = \frac{5}{9}$$

Check by listing: $B \cup C = \{1, 6, 7, 8, 9\}$ has 5 elements → $\frac{5}{9}$ ✓. Without subtracting $\frac{2}{9}$, the 7 and 8 would be counted twice.

Venn diagrams:

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A and B (mutually exclusive)          B and C (overlapping)

 ┌─────────┐   ┌─────────┐          ┌─────────┬─────┬─────────┐
 │1 2 3 4  │   │  6 7 8  │          │    6    │7 8  │  1  9   │
 └─────────┘   └─────────┘          └─────────┴─────┴─────────┘
      A             B                  B only   BnC   C only
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In the left diagram the circles don't touch, so $P(A \cup B)$ is just $\frac{4}{9} + \frac{3}{9}$. In the right diagram 7 and 8 sit in both circles. Adding $\frac{3}{9}$ and $\frac{4}{9}$ would count them twice, so $P(B \cap C) = \frac{2}{9}$ is subtracted once.

> **Word cue:** "**or**" → ∪ → **addition** rule. "**and**" → ∩ → **multiplication** rule.

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4. Joint probability - $P(A \cap B)$ (multiplication rule)

$P(A \cap B)$ is read "the probability of A **and** B" (both happen):

$$\begin{aligned} \text{Words:} \quad & P(A \text{ and } B) = P(A \text{ given } B)\, P(B) = P(B \text{ given } A)\, P(A) \\ \text{Symbols:} \quad & P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A) \end{aligned}$$

Independent events (one does not affect the other): $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$, so

$$\begin{aligned} \text{Words:} \quad & P(A \text{ and } B) = P(A)\, P(B) \\ \text{Symbols:} \quad & P(A \cap B) = P(A)\, P(B) \end{aligned}$$

Independent or dependent?

Situation	Type	Why
Draw a red marble, **put it back**, then draw a blue one	**Independent**	Replacing the marble restores the bag
Draw a red marble, **keep it**, then draw a blue one	**Dependent**	The bag now has one fewer marble, so the second probability changes
Toss a coin twice	**Independent**	The first toss doesn't affect the second

Notation warning: $P(A \cap B)$ vs $P(AB)$

- $P(A \cap B) = P(B \cap A)$: both happen, no order implied.
- Some books write $P(AB)$ for "A **first, then** B" and $P(BA)$ for "B **first, then** A". These are **not always equal**.
- Read the context: are the events **simultaneous**, or does one come **before** the other?

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5. Conditional probability - "A given B"

$P(A \mid B)$ is the probability of A **given that** B **has already happened**:

$$\begin{aligned} \text{Words:} \quad & P(A \text{ given } B) = \frac{P(A \text{ and } B)}{P(B)}, \qquad P(B \text{ given } A) = \frac{P(A \text{ and } B)}{P(A)} \\ \text{Symbols:} \quad & P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B \mid A) = \frac{P(A \cap B)}{P(A)} \end{aligned}$$

Conditional = joint ÷ marginal.

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6. Bayes' theorem

Replacing $P(A \cap B)$ with $P(B \mid A)\,P(A)$ from the multiplication rule gives

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$$

It is the conditional probability formula with the joint probability rewritten.

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7. Negation (complement) probability

$$\begin{aligned} \text{Words:} \quad & P(\text{not } A) = 1 - P(A) \\ \text{Symbols:} \quad & P(A') = 1 - P(A) \end{aligned}$$

$P(A')$ is read "the probability of **not** A". The complement is also written A^c (the most common form in textbooks), $\bar{A}$ or $\sim A$.

- $P(A') = 1 - \frac{4}{9} = \frac{5}{9}$. Check: S has 5 elements outside A (5, 6, 7, 8, 9) ✓
- If $P(B) = 0.35$, then $P(B') = 1 - 0.35 = 0.65$, a 65% chance that B does not occur.

This works because the total probability of all outcomes is 1.

Formula summary

Type	Words	Symbols
Marginal	$P(A) = \frac{\text{favorable}}{\text{total}}$	(same)
Union (or)	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Union, mutually exclusive	$P(A \text{ or } B) = P(A) + P(B)$	$P(A \cup B) = P(A) + P(B)$
Joint (and)	$P(A \text{ and } B) = P(A \text{ given } B)P(B) = P(B \text{ given } A)P(A)$	$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$
Joint, independent	$P(A \text{ and } B) = P(A)P(B)$	$P(A \cap B) = P(A)P(B)$
Conditional	$P(A \text{ given } B) = \frac{P(A \text{ and } B)}{P(B)}$	$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
Bayes	$P(A \text{ given } B) = \frac{P(B \text{ given } A)P(A)}{P(B)}$	$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$
Complement	$P(\text{not } A) = 1 - P(A)$	$P(A') = 1 - P(A)$